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Bachelor of Engineering

Application of Transformer neural network on sea surface temperature prediction

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Abstract

Sea Surface Temperature (SST) prediction is crucial for understanding and forecasting climate change, weather patterns, and their ecological impacts. However, the inherent complex spatio-temporal dependencies make accurate prediction challenging. To achieve higher accuracy and efficiency in SST forecasting, a deep learning model based on self-attention mechanisms, hereinafter referred to as 'SSTmodel', has been constructed. This model preprocesses data using multi-scale convolutional networks to capture spatial features and employs self-attention layers to process time series data, effectively capturing long-term dependencies and enhancing the model's response ability and predictive accuracy to dynamic changes in SST. The experiment results demonstrate that the model adapts well to both global and local regions, with a bias of approximately ±0.001°C and RMSE not exceeding 0.24°C, across 3 regions, consistently outperforming traditional ConvLSTM and convolutional network-based DL models in terms of efficiency and accuracy. Additionally, the model shows high accuracy in long-term prediction and the effective identification of the occurrence of occurrence of El Niño and La Niña phenomena. Finally, an interactive visualization interface has been constructed to intuitively reflect the dynamic changes in SST distribution.

Keywords: Sea Surface Temperature prediction, spatio-temporal sequence , self-attention, convolutional networks, dynamic changes, long-term prediction, El Niño phenomena

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1 Introduction

1.1 Research value and significance of SST prediction

As is known to all, the oceans account for 71% of the earth's surface area, and there is a constant direct exchange of heat and moisture between the ocean and the atmosphere [1, 2], which in turn influences wind direction, wind speed, atmospheric circulation, and precipitation distribution [3, 4]. Variations in sea surface temperature (SST) significantly impact global climate, marine ecosystems (such as underwater algae and coral reef ecosystems [5]), and marine resource management [1], thus greatly affecting human living conditions and productive activities. Moreover, SST serves as a critical indicator of oceanic climatic phenomena such as El Niño and La Niña. As a primary source of interannual climate variability on a global scale, the El Niño-Southern Oscillation (ENSO) is characterized by anomalously warm surface water temperatures in the central and eastern tropical Pacific, occurring every few years and lasting from a few months to two years [6]. This phenomenon typically triggers extreme weather events worldwide, including hurricanes, heatwaves, and droughts [1, 2, 5]. The high temperature of seawater is one of the primary drivers for the development of severe storms, and temperature distribution is closely related to the formation, intensity, and path of hurricanes [7]. Furthermore, under the influence of global warming, the occurrences of La Niña and El Niño have become more frequent since the 1960s. Accurate prediction of sea temperatures can enhance the accuracy of weather forecasts, improve disaster warning capabilities, and formulate effective climate change adaptation strategies, thus playing a vital role in various sectors such as agriculture, fisheries, and maritime operations [8, 9], with potential significant economic and social benefits.

1.2 Related research process and current situation

It is noteworthy that due to the quasi-periodic nature of climate variations often exhibited by SST, achieving high accuracy in predictions is both feasible and highly anticipated [1]. Consequently, over the past few decades, researchers have made considerable efforts and achieved a series of technical advancements in understanding and predicting SST variations and the El Niño phenomenon [1, 10].

From the perspective of oceanic data collection technology, there has been a significant leap from buoy-based to satellite remote sensing. Initially, SST measurements relied on buoy and ship observations, thus limiting the coverage area. Even with the use of automated observing equipment like Argo floats [Wong et al., 2020], spatial coverage remained uneven and the data collection intervals were lengthy. With advancements of remote sensing

technology, satellites have become the primary means for monitoring SST, providing broader and more continuous data, significantly enhancing the resolution and coverage of SST data collection. Although electromagnetic radiation does not penetrate deeply into seawater, it is possible to infer sub-surface data based on surface parameters like SST and surface height anomaly (SLA) through a combination of dynamical models and field observations [Kao, 1987; Wunsch and Gaposchkin, 1980] or purely statistical relationships [Fiedler, 1988]. Chu et al. [1997] developed a parameterized model and established the inversion of the Ocean Subsurface Thermal Structure (OSTS) [2000] [4, 8].

At present, there are two primary methods for applying satellite remote sensing imagery to sea temperature forecasting. The first method is the dynamic approach, which involves creating numerical models based on mathematical principles and physical equations to simulate the Earth's climate system and predict changes in sea temperature. This method relies on mathematical equations and knowledge of ocean dynamics to simulate the spatio-temporal evolution of ocean temperatures [4, 8], and typically depend on complex mathematical models and fundamental physical equations of the climate system, such as fluid dynamics and General Circulation Models (GCMs) [7]. Lapeyre and Klein (2006) developed an effective Surface Quasi-Geostrophic (e-SQG) model that uses surface density information to reconstruct the three-dimensional velocity field in the upper ocean, with their findings validated through simulations of the dynamics of the Antarctic Circumpolar Current [19]. This approach provides a deep understanding and explanation of various complex physical processes, and the model outputs usually possess high scientific accuracy. However, it also requires high-performance computing resources and a corresponding knowledge base, and depends on the accuracy of initial conditions (input parameters) and parameterization schemes [2].

The data-driven approach, which focuses on using statistical or machine learning techniques to learn underlying relationships and patterns from historical data to predict future changes in ocean temperatures [2, 4]. Starting from the late 1990s, the evolution from simple neural networks and statistical models to the application of complex machine learning algorithms such as Support Vector Machines (SVM) and Random Forests has gradually been incorporated into global climate model analysis tools. Entering the 2010s, with the widespread application of deep learning technologies, particularly Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), there has been a significant improvement in SST prediction accuracy and time series processing capabilities. For example, in 2018, Yang and others significantly enhanced the accuracy of SST predictions using the

CFCC-LSTM model [7]. Zheng and colleagues (2020) developed a deep learning model based on satellite data, successfully predicting the SST evolution of tropical instability waves, demonstrating the potential of data-driven methods in predicting marine phenomena [11]. Although data-driven approaches excel in processing efficiency and flexibility, they are highly dependent on data quality and often lack sufficient physical process explanations.

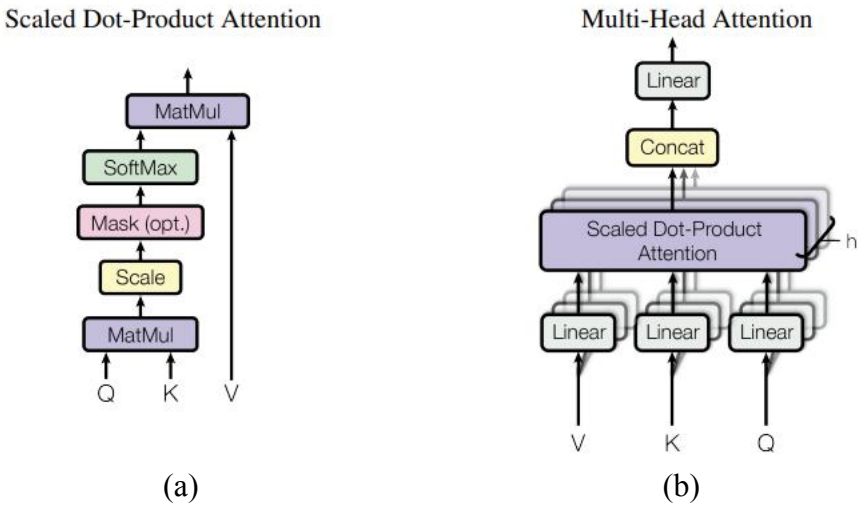
There are also instances where both methods are combined, such as in 2004 when M. M. Ali and others used a neural network approach to estimate subsurface ocean temperature structures, enhancing the model's accuracy in predicting deep-sea temperature distribution by integrating dynamic height and sea surface temperature data [8]. Furthermore, satellite-based Earth observation and marine remote sensing technologies continually enhance their performance through deep learning, supporting more accurate data collection and analysis [4].

1.3 The advantages of the Transformer

1.3.1 Self-Attention Mechanism

The self-attention mechanism of the Transformer allows the model to focus on important parts of the input sequence and understand the relationships between these parts, thereby effectively handling complex patterns in text and images.The relevant formula and structure are as follows, see [12] for details.

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



(a) (b)

Figure 1-1 (a) Scaled Dot-Product Attention; (b) Multi_Head Attention

1.3.2 Scalability and Parallelization

Du to its structural design [12], the Transformer model can process data in parallel across multiple processing units, significantly speeding up the learning process. Moreover, the model can flexibly adapt to tasks of varying complexity by increasing the number of layers or adjusting its size.

1.3.3 Applications Across Domains

- NLP: used for text classification, machine translation, etc.
- CV： Applied to image classification, object detection, etc.
- Cross-mode: used for image caption generation, visual quiz, etc.

Related to SST: In meteorological prediction applications, Huawei Pangu and Nvidia FourCastNet with the best performance (the highest prediction accuracy) all take Transformer as an important architecture [12, 13, 14].

1.4 Excellent research based on Transformer in this field

1.4.1 3D-Geoformer model

The 3D-Geoformer model is specifically developed for multivariate predictions in Earth sciences, primarily focused on forecasting the El Niño-Southern Oscillation (ENSO). The model employs an innovative self-attention mechanism to capture complex spatiotemporal correlations and performs rolling predictions by simulating dynamic coupled models. It demonstrates high accuracy in long-term predictions, particularly in predicting SST anomalies related to ENSO. [8, 10]

In terms of predictive performance, it excels in forecasting the cooling conditions in the tropical Pacific for the years 2020-2022. The model accurately predicted the double La Niña states in 2020 and 2021, closely matching observed values and outperforming average predictions collected by the International Research Institute (IRI) and the North American Multi-Model Ensemble (NMME). The 3D-Geoformer, trained through its self-attention mechanism, effectively handles complex multivariate relationships, especially the intricate interactions between wind stress and ocean temperature anomalies, which are key factors influencing El Niño phenomena. For instance, it successfully captures the evolving relationships between wind stress, SST, and subsurface temperature during the second-year cooling conditions in 2021. [8, 10].

1.4.2 TemproNet model

TemproNet is a deep learning model that combines transformers and convolutional neural networks. This model utilizes multivariate sea surface observational data, including Sea Level Anomaly (SLA), Sea Surface Temperature (SST), and Sea Surface Wind (SSW) as inputs. It features hierarchical transformer encoders to extract high-level features and employs a lightweight convolutional decoder to extract low-level features. This is further refined through a four-stage encoder process to ensure the capture of temperature information from the ocean surface down to deeper levels [4].

In the prediction of 32 temperature profiles at a depth of 200 meters in the South China Sea, TemproNet demonstrated superior performance with a Root Mean Square Error (RMSE) of 0.52°C and a coefficient of determination (R^2) of 0.83. These metrics are significantly better than those of other models such as Attention-Unet, LightGBM, and ANN. Additionally, TemproNet not only excels in regional applications but also has the potential to be widely applied in other global marine areas [4].

1.4.3 Instance conclusion

The application of these two Transformer-based models in sea surface temperature (SST) prediction highlights the enormous potential of Transformer technology in handling complex Earth science data, especially in addressing the intricate spatiotemporal dynamics of ocean and atmospheric interactions. The 3D-Geoformer excels in simulating long-term dependencies and non-local connections, while TemproNet demonstrates its strengths in accurately predicting regional sea temperature variations. Both models emphasize and exemplify the adaptability and flexibility of transformers, which enable the integration of diverse data sources, thereby enhancing the potential for global or regional climate prediction applications.

2 Implementation

2.1 Dataset selection

The dataset used in this project is derived from temperature values within the geographical range of [120°W, 180°W] and [-10°N, 10°N], as illustrated in Figure 2-1(a). This region is predominantly covered by the ocean and is situated in the central-eastern part of the Pacific Ocean. The dataset comprises sea surface temperature values for each day from June 1, 2002, to May 31, 2018, sampled at regular intervals. As depicted in Figure 2-1(b), 'samples' represents the collection of all these values, totaling 496*264*5889 data points. Here, 5889 represents the number of samples (i.e., days), with each element representing the sea surface temperature at a specific spatial location at a specific time point. Calculations reveal that the dataset has a spatial resolution of 0.12°(in longitude) and 0.0758° (in latitude), respectively, offering higher resolution compared to most similar datasets [11]. The effects generated by the low resolution can be referred to in the [10]. In Python, temperature distribution plots corresponding to these values can be generated using seaborn.

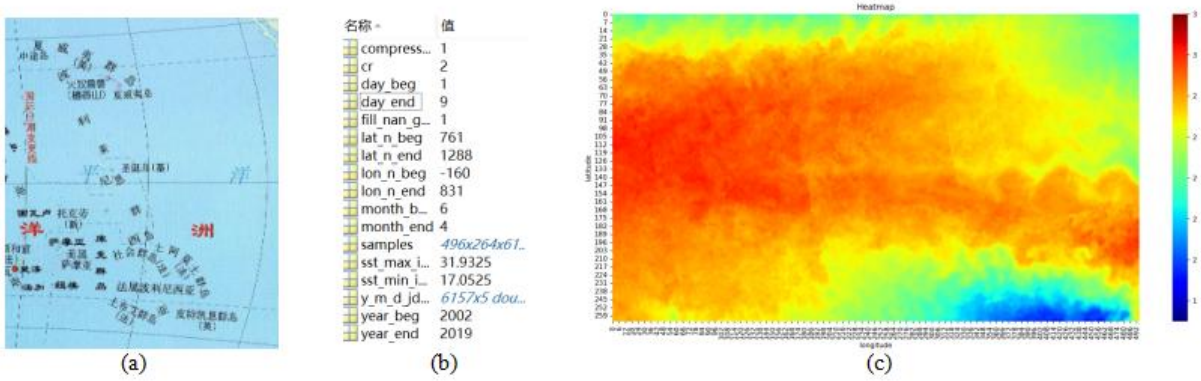


Figure 2-1 The Data set source. Collection and representation. (a) The actual geographic region corresponding to the data set; (b) The numerical data collected; (c) Temperature distribution map constructed from data

The region located in the central-eastern equatorial Pacific, which encompasses the data collection interval, is known as the most frequent occurrence zone of the El Niño and Southern Oscillation (ENSO) phenomenon [7, 10]. It is characterized by prolonged and anomalous warming, with its impact extending globally through ocean-atmosphere interactions [10]. As referenced in Section 1.1, this area serves as a crucial zone for observing and monitoring ENSO, and its significance extends beyond sea surface temperature (SST) anomalies.

2.2 Dataset Processing

This project will establish a sea surface temperature prediction model based on the Transformer architecture. To explore its prediction effectiveness both globally and regionally, the dataset will be split into cropped areas of 64*64 (region 1, maintaining the original spatial resolution), 128*128 (region 2, obtained by bilinear interpolation downsampling from 256*256), and 128*128 (region 3, obtained through bilinear interpolation from a full global area of 496*256). In this project, the decision to downsample the global area from 496x256 to 128x128 was primarily made to enhance computational efficiency and reduce memory requirements, while ensuring that sufficient and comprehensive information is available for model training.

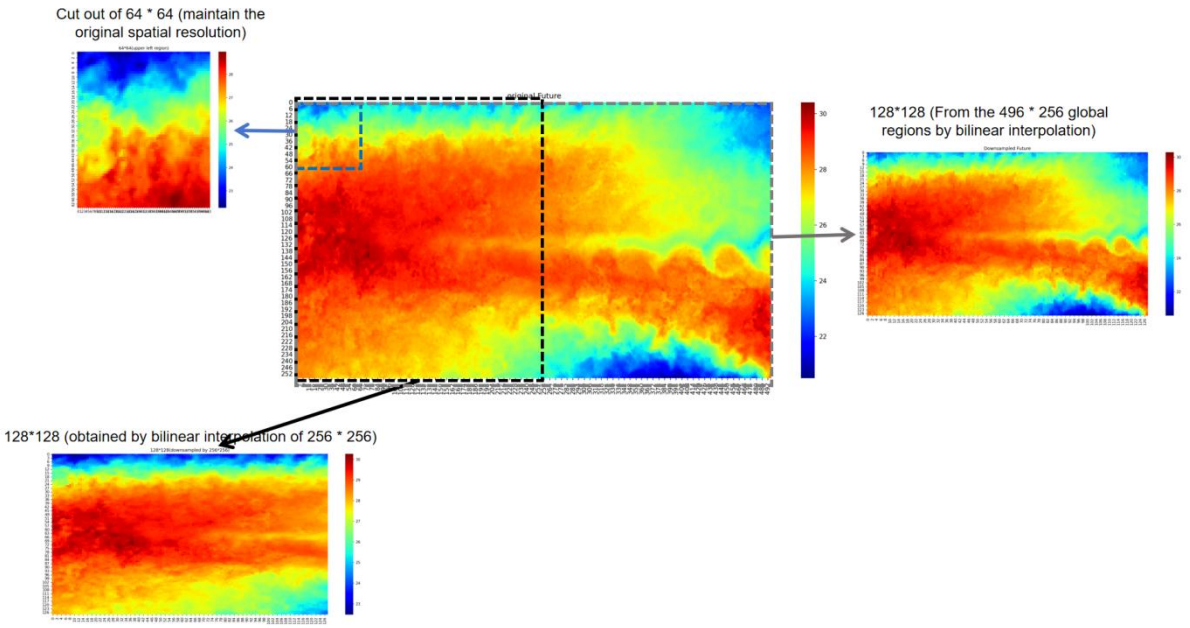


Figure 2-2 The 3 regions obtained after splitting and processing the dataset

The use of bilinear interpolation in this project is primarily aimed at maintaining image quality while improving computational efficiency. Specifically, this method estimates the value of new pixels by calculating the weighted average of the four adjacent pixels in the original image, effectively preserving the edges and details of the image. It considers the spatial relationships between pixels, avoiding the abrupt pixel value changes that might occur with simple averaging, thereby delivering a clearer and smoother image output. Additionally, since the original dimensions and the target dimensions in this case are not exact multiples of each other, bilinear interpolation better handles this non-divisible scaling issue, ensuring that image quality and spatial features are preserved during the downsampling process. This provides high-quality data for model training. In contrast, simple average downsampling,

though perhaps more straightforward computationally, does not preserve image details and quality as effectively as bilinear interpolation. [15, 16]

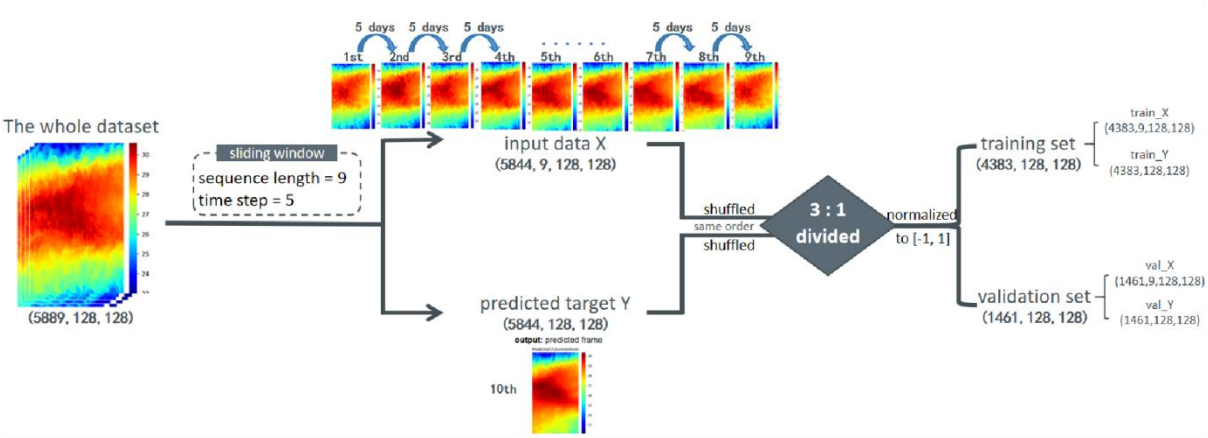


Figure 2-3 The subsequent processing flow of 3 regions

In terms of following dataset processing, a "sliding window" technique is first used to construct the input data X and the target data Y. The window length is set to 9, and the time step is chosen as 5 instead of 1 to ensure data processing efficiency while reducing the risk of model overfitting and improving performance on unseen data. This means that it is using 9 frames collected every 5 days within a continuous 45 days to predict the target image on the 50th day, followed by a series of processing such as shuffling and normalization.

To be specific, each input data X will contain 9 frames (each frame spaced 5 days apart) of temperature matrices, with the subsequent frame's real temperature data used as the output data Y, which is the expected target. For example, if the input data starts on June 1, 2002, as the first frame, the second frame would correspond to June 6, 2002, the third to June 11, 2002, and so on, the ninth frame would be extracted from June 11, 2002, with the target frame being the SST distribution on June 16, 2002. In entire training process, the shapes of X and Y obtained are (5844, 9, 64, 64) and (5844, 64, 64) respectively for 5844 day forecasts. Subsequently, X and Y are shuffled and divided into a training set and a validation set in a 3:1 ratio. Validation on this dataset has shown that a random arrangement, as opposed to a chronological order, can further suppress overfitting and enhance generalization capability. After partitioning, normalization is applied to adjust the numerical range to between [-1, 1], which helps improve the efficiency and stability of model training.

2.3 Model constructing

For this spatiotemporal sequence prediction task, the model takes 9 SST distribution maps as input and the shape is denoted as (T, W, H), where T equals 9 for the number of time steps, and W and H correspond to the width and height of the SST distribution matrix,

respectively. The constructed model structure, as shown in figure 2-4, effectively integrates multiple neural network technologies to optimize the accuracy and efficiency of sea surface temperature (SST) prediction. By incorporating convolutional neural networks (CNNs) in preprocessing part and deconvolution layers in output part, the model can efficiently extract and reconstruct spatial features; Meanwhile, by emphasizing spatial and temporal encodings and effectively applying multi-head attention in the Transformer encoding layers, the model not only processes traditional one-dimensional sequence data but also effectively manages and predicts multi-dimensional time series data. In the post-processing stage, 3 bias correction maps were proposed and added to correct the systematic bias. The design and optimization of these structures not only enhance the model's performance but also significantly reduce the number of parameters needed, lower the demand on hardware memory, and save computational resources. Next, region 1 will be used as an example, and the complete model architecture is described along the detailed model prediction implementation process and compared to the traditional Transformer prediction model[14] at the same time:

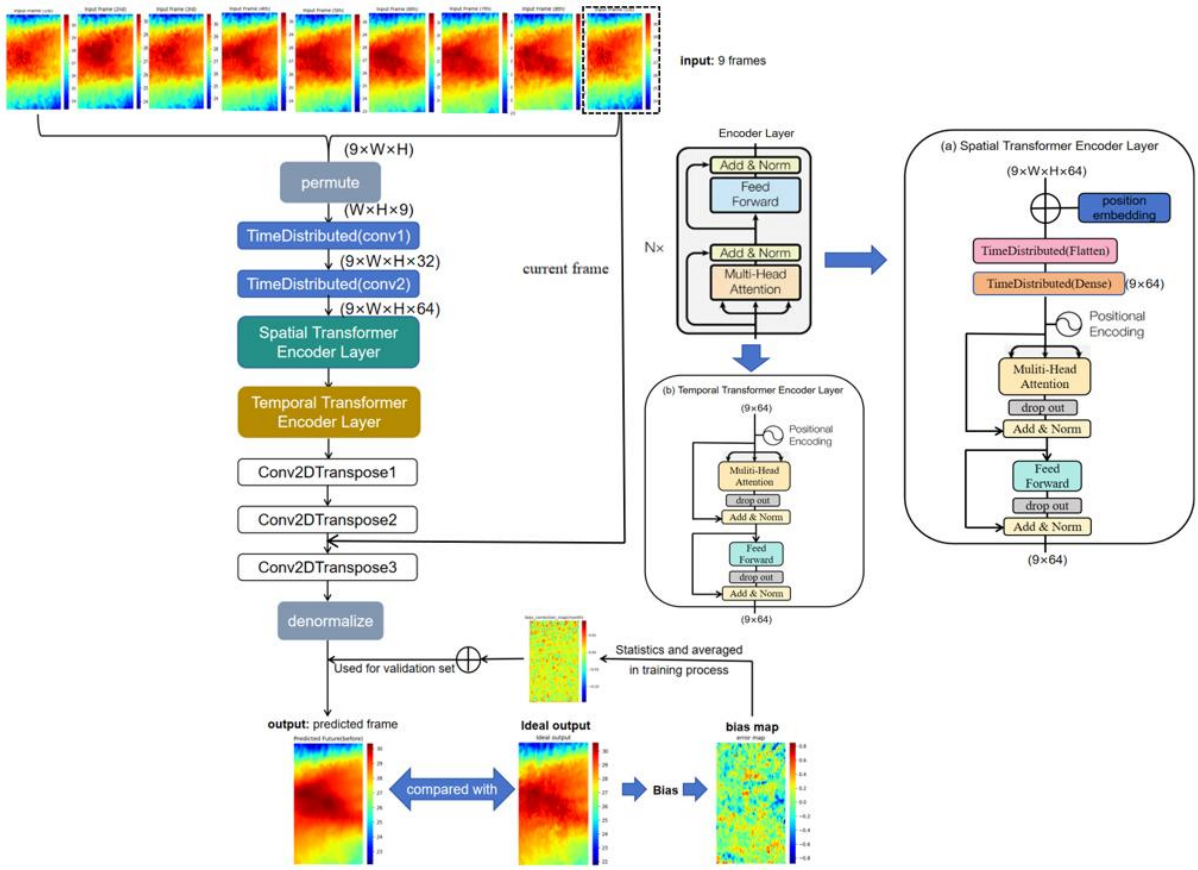


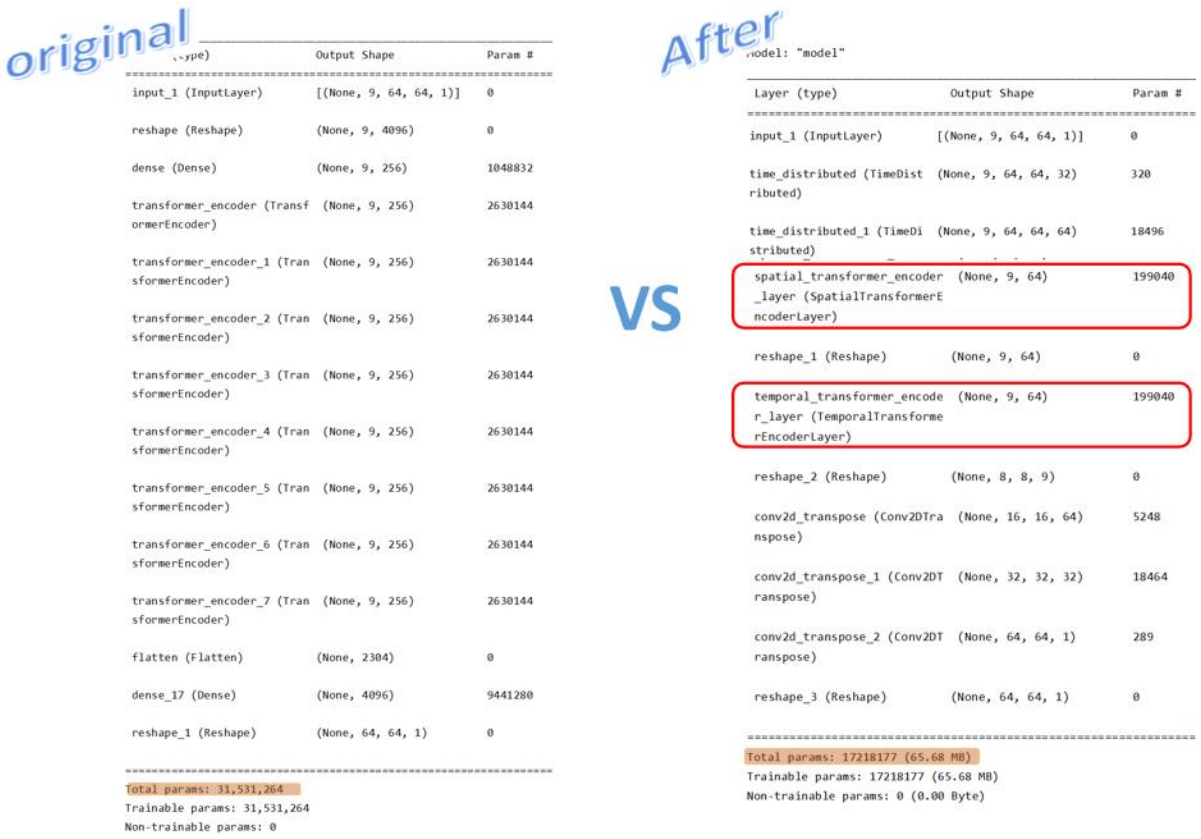
Figure 2-4 ‘SSTmodel’ model structure

2.3.1 Multi-layer convolution processing for input images

Firstly, considering the multi-dimensional input of spatio-temporal sequences, the model initially employs a TimeDistributed layer paired with Conv2D for convolutional processing of the input SST time-series images. TimeDistributed ensures that the same operations are applied independently at each time step, maintaining consistency in the processing of each frame in the sequence. The Conv2D layers extract spatial features from each frame, with the initial layers utilizing 32 filters and subsequent layers using 64 filters, all with a kernel size of 3x3, to progressively capture more complex spatial information. Additionally, the use of padding='same' maintains the dimensional integrity of the images, while the application of L2 regularization helps reduce overfitting, ensuring the model's generalization capability. The use of the ReLU activation function introduces non-linearity, enhancing the model's ability to capture complex data relationships. This structural design not only optimizes feature extraction efficiency but also ensures effective processing of time-series data [17, 18, 19].

2.3.2 The builted Temproal-Spatial Trnsformer Encoders

In the processing of spatio-temporal sequence data such as Sea Surface Temperature (SST), the traditional Transformer architecture, with its multiple encoder layers, may not be sufficient to effectively distinguish and manage the spatial and temporal information within the time series. Thus, as figure 2-5, a significant improvement has been made by subdividing and modifying the multiple encoder layers into a single spatial encoder layer 'Spatial Transformer Encoder Layer', and a temporal encoder layer 'Temporal Transformer Encoder Layer'. Their specific structure is shown in figure 2-6 and this approach allows the model to specialize in handling spatial and temporal information, thereby enhancing processing efficiency and predictive accuracy [19, 20].



(a) (b)

Figure 2-5 The model structure print diagram (a)A traditional Transfomer structure used for prediction ;

(b) 'SSTmodel' structure

The spatial encoder layer focuses on analyzing critical spatial information at each time step. For example, it extracts vital spatial features from SST images, such as temperature gradients or the temperature distribution of specific regions. This processing ensures that the model comprehends and captures the spatial structure and local features within each frame. Meanwhile, the temporal encoder layer concentrates on the evolution of features over time, analyzing how spatial features from consecutive frames change over time. This is crucial for predicting future states, as the dynamics of temperature changes often depend on previous trends.

The clear separation of spatial and temporal processing ensures that the model can learn and understand the complex spatial and temporal dependencies present in SST data more profoundly. It enhances the model's performance in handling dynamic changes. Moreover, such separation allows for more flexible adjustments to the model architecture and optimizes the allocation of computational resources.

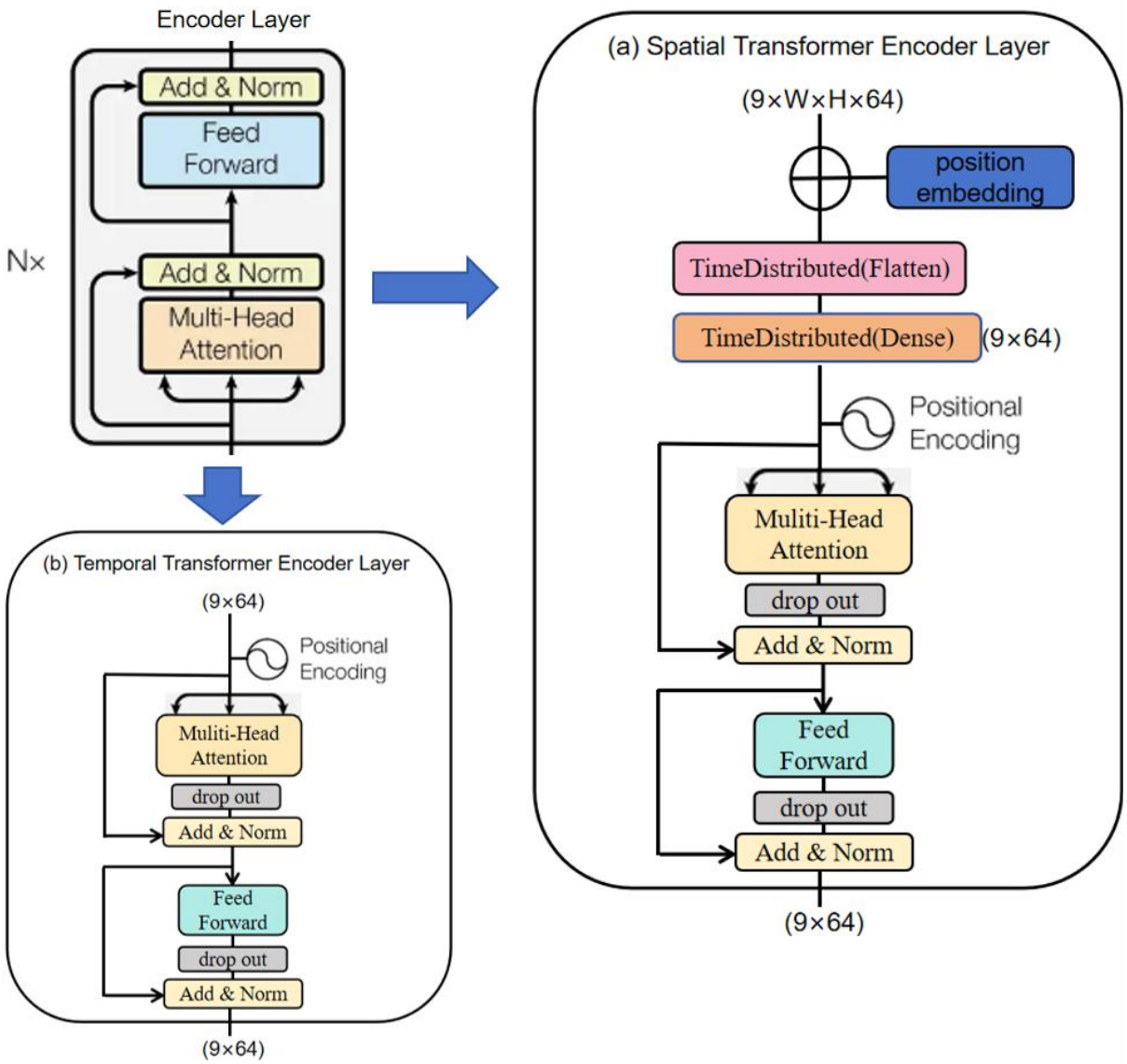


Figure 2-6 The traditional Transformer Encoder layer and (a)Spatial Transforme Encoder Layer ; (b) Temporal Transformer Encoder Layer

As shown in figure 2-6(a), the structure of the modified 'Spatial Transformer Encoder Layer' consists of adding a Position Embedding, followed by a Dense layer and Position Encoding, ahead of a traditional Transformer Encoder layer. This design is based on the specific functions and purposes of each component in processing data.

Additionally, positional information is crucial in the traditional Transformer architecture. This is because the self-attention mechanism is inherently position-invariant, meaning that without some way to account for the positions of elements within the input sequence, the model would be unable to distinguish the order of elements. To address this issue, Position Embedding and Positional Encoding are introduced sequentially here. After the previous multi-layer convolution processing, the purpose of adding Position Embedding is to inject spatial positional information into the model and provide a learnable and optimizable space to

better adapt to the positional demands of this specific tasks. At the initial stages of image processing, by assigning a unique position embedding to each pixel block at every spatial location within each time step, the model begins to understand the position of each element within the overall data structure. Moreover, this 'positional embedding' operation is dynamic, which can be adjusted and optimized through the training process to best suit the underlying inference logic of SST image prediction.

However, standalone positional embeddings may still fail to fully express all possible interactions and complex features of these positions. The introduction of fully connected layers (Dense layers) is intended to further process and transform these input data that already contain positional information. Dense layers can learn the nonlinear relationships between different positions, enhance the features in positional embeddings, and generate more complex and abstract feature representations. This transformation is particularly useful for image data characterized by high nonlinearity and complex spatial relationships. Additionally, in the context of attention mechanisms, the complexity is linearly related to the square of the input length and the dimension of features. The Dense layer transforms the data from (9, W×H×64) to a more compact (9, 64), significantly reducing the number of parameters and the computational complexity required to process these data (approximately to 1/2 of the original). This not only helps save memory but also potentially enhances the efficiency and speed of the model's operation.

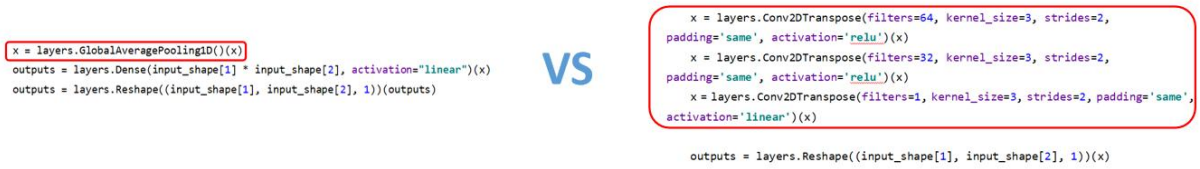
Unlike positional embeddings, the subsequently added Positional Encoding is fixed and is introduced to bring the model's attention mechanism into specific time series analysis. By introducing positional encoding after enhancing spatial features with Position Embedding, the model is not only capable of understanding complex spatial relationships but also able to analyze these relationships across the temporal dimension, which is essential for time series data. Furthermore, this ensures that the model remains sensitive to positional information even without further learnable location parameters, compensating for positional features that may not be fully optimized during training.

Position Embedding provides foundational, learnable location information; the Dense layer enhances the model's ability to learn complex data patterns; while Positional Encoding ensures the comprehensiveness of location information and the coherence of the model in handling sequence tasks. Such a design allows the model to not only have a deeper understanding of positional dependencies within the data but also handle more complex and abstract tasks, improving overall performance and adaptability. This strategy is particularly

suitable for SST prediction application that have a high sensitivity to location and meet the requirement of more specific expressions for element position.

2.3.3 Multi-layer transposed convolution used in output part

In the output section, multiple layers of transposed convolutional layers (sometimes referred to as deconvolutional layers) are employed to progressively upsample and restore the original high resolution, effectively reconstructing detailed spatial features of sea surface temperature (SST). This approach significantly enhances the mitigation of image homogenization issues and improves the extraction of contours and textures. For region 1, three transposed convolutional layers are used, all with a kernel size of 3x3 and filter counts of 64, 32, and 1 respectively, with a stride of 2. For region two and the global image, there are five transposed convolutional layers with kernel sizes of 3x3 and filter counts of 64, 32, 16, 8, and 1, with strides of 2, 2, 2, 2, and 1 respectively. Given that this is essentially a regression problem, the predicted temperature distribution values should correspond to actual temperature values, which may include negative numbers or values beyond a specific range. The activation function of the last transposed convolutional layer is 'linear', ensuring that the output is not distorted by nonlinear compression or stretching. Using a ReLU activation function, which sets all negative outputs to zero, is unacceptable in situations where accurate prediction of positive and negative values is required. Furthermore, following the structures referenced in [12, 21], the temporal sequence of events is crucial for time series prediction. Thus, the output from the penultimate transposed convolutional layer is merged with the input of the ninth frame (current frame) and fed into the final transposed convolutional layer. While this processing does not show advantages in the 64x64 region 1, when expanded to the 128x128 of region 2 and the global image application, it results in a reduction of the Mean Squared Error (MSE) by around 0.005°C².



(a) (b)

Figure 2-7 (a) Average pooling plus the full connection layer; (b) Multilayer transposition convolution

Homogenization issues typically refer to when models processing image or spatial data produce outputs that are overly smooth, lacking detailed textures and clear boundaries. As visible from Figure 2-7(a), when the output layer employs average pooling followed by a Dense linear layer, it pools the feature map into a one-dimensional vector, losing most of the

spatial information, simplifying the complexity of spatio-temporal data, and leading to homogenized prediction results in Figure 2-8(a). Conversely, the use of multiple layers of convolutional transpose layers (Figure 2-8(b)) significantly improves contour extraction and greatly reduces the homogenization issue [12, 22, 23]. This is due to its unique working mechanism (interpolation followed by convolution), which more naturally restores the spatial structure and details of the image. Compared to traditional upsampling methods (such as direct interpolation) and average pooling with a Dense linear layer, the transpose convolution effectively reconstructs contours and texture information in the image while upsampling. This is achieved by adjusting and optimizing the contextual information around each pixel through the weights learned in the filters. This not only involves converting low-resolution feature maps into high-resolution outputs step by step, but also leverages the benefits of local connectivity and weight sharing at each step of upsampling, and can focus on restoring detailed features in specific areas of the image. Therefore, transpose convolution layers do more than merely expand the spatial dimensions of an image; they enhance the learning and expression of key visual features during the reconstruction process, such as edges, corners, and complex texture patterns. The experiment result comparison reflect the optimization effect of the predicted image reconstruction, table 3-1 presents the result comparison between this two output structure in the incomplete model during model constructing process. Obviously, MSE and Bias has been reduced to 0.13°C² and -0.00597°C.

Table 3-1 Experiment result comparison of the two output layer structures

	Multilayer transposition convolution	Average pooling plus the full connection layer
MSE (°C²)	0.13135	0.57798
Bias (℃)	-0.00597	-0.00273
RMSE (℃)	0.36199	0.24031
MAE (℃)	0.28035	0.18488

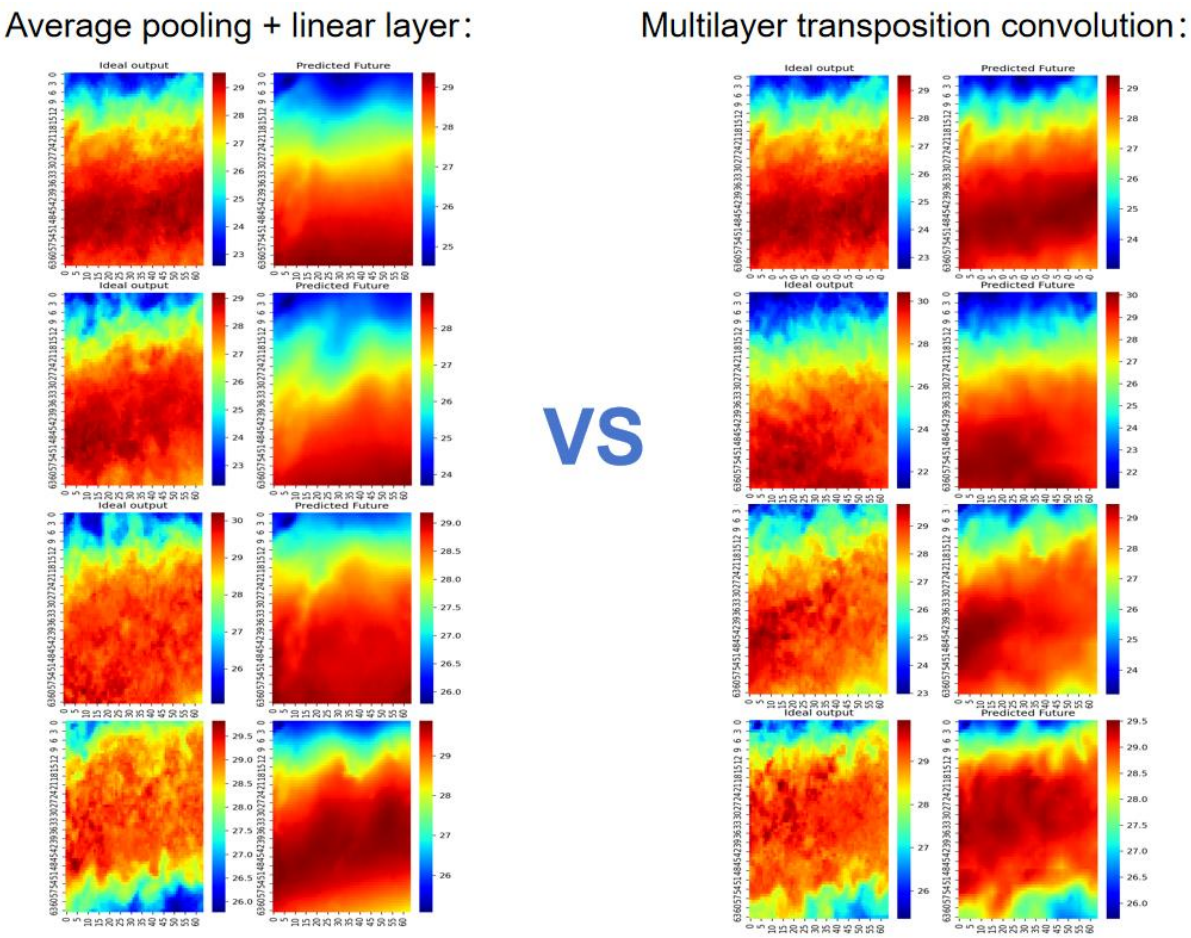


Figure 2-8 The output of the predicted graphs of two output layers (a) The outputs when using the Average pooling plus the full connection layer; (b) The outputs when using the Multilayer transposition convolution

Table 3-2 Comparison of the two output layer structures

	Multilayer transposition convolution	Average pooling plus the full connection layer
Retention of the spatial information	Can maintain the spatial structure of the image by learning the details in the reconstruction process	Easy to substantial loss of spatial information because pooling simplifies the local features of the data
computational efficiency	Reducing the number of parameters of the model and improving the computational efficiency through local connectivity and parameter	When processing the data of large-size images, the number of parameters is large and the computational cost is high

	sharing	
Details recovery ability	Especially suited for recovering image details such as edges and textures as it allows more refined control over the upsampling process	The Dense layer is often difficult to reconstruct these details precisely, and the output images may appear more ambiguous

2.3.4 Adding bias correction map at post-processing stage

Even high-performance models may exhibit systematic biases during prediction due to limitations in training data, insufficient features, or overgeneralization. Therefore, referencing the post-processing stage from [11], the average bias correction map, referred to as Bias Map 1 in Figure 2-9, is computed by averaging the deviations between each prediction result and the ideal output during the training process. This bias is then graphically applied to future predictions, effectively correcting these deviations and enhancing the overall prediction accuracy of the model. Considering the seasonal and periodic variations in SST distribution [11, 12], two new error correction maps were statistically derived based on the training dataset.

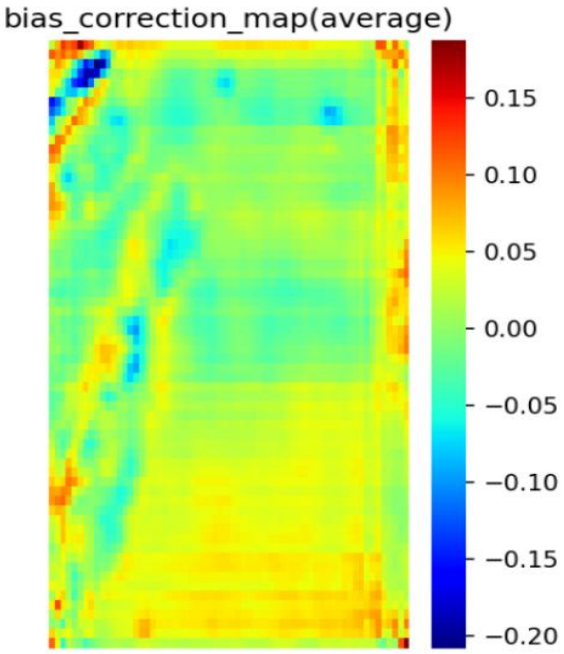


Figure 2-9 Bias Map 1

Bias Correction Map 2: Considering seasonality, the analysis is refined to each month here [11]. The average deviation maps for each month over 12 years of training data have

been statistically analyzed, with results presented in Figure 2-10. The deviation distribution varies significantly by month, especially in October, November, and December, where the maximum deviation reaches up to 2°C. The blue line in Figure 2-11 represents the average monthly deviations, which range between (-0.232°C, -1.815°C). After subtracting the overall average deviation correction map, the resulting orange line in Figure 2-11, although reduced to (-0.975°C, 0.515°C), still shows significant and unevenly distributed deviations. Therefore, the introduction of a corresponding monthly deviation correction map (Figure 2-10) is warranted.

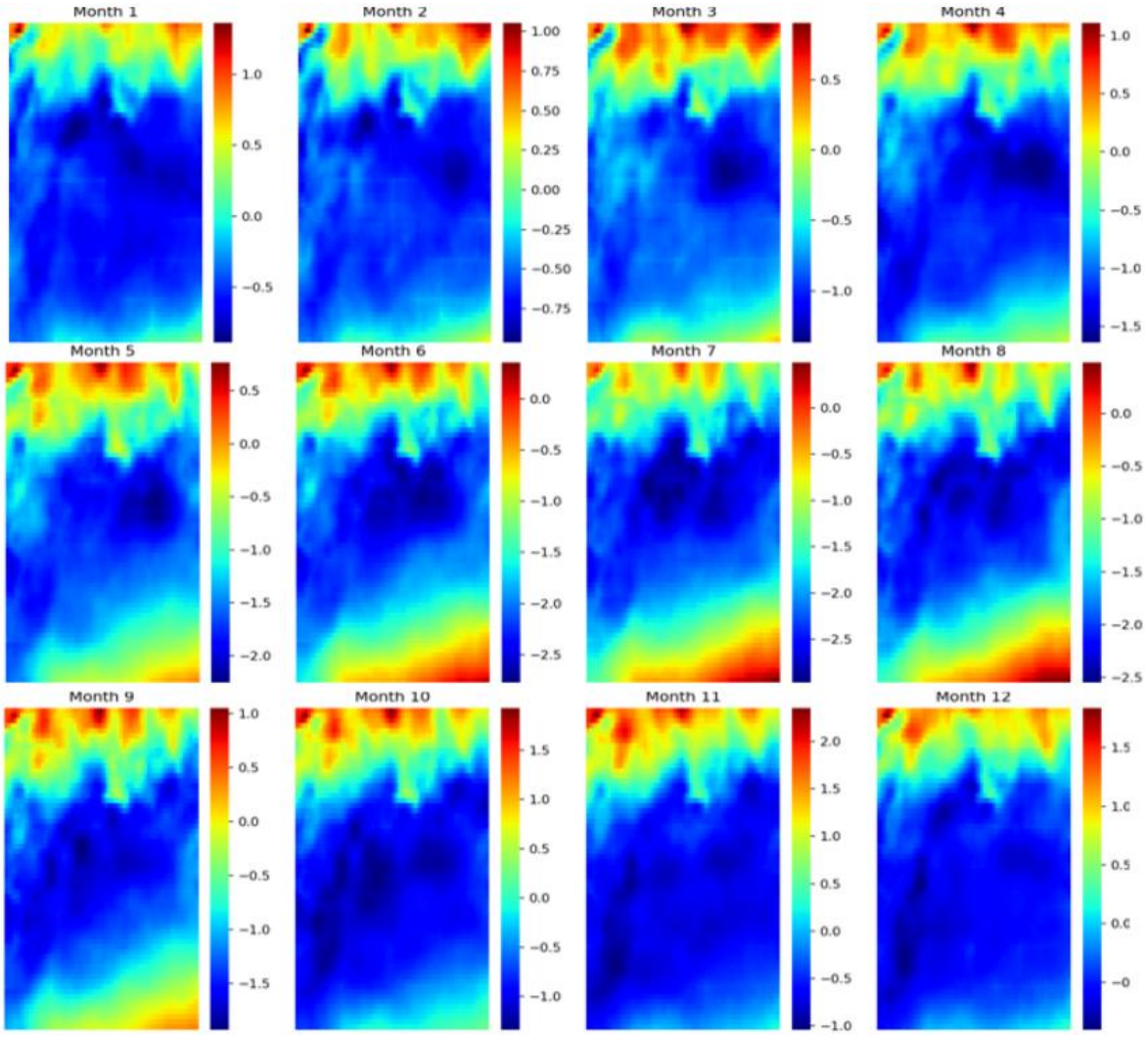


Figure 2-10 Monthly deviation correction map

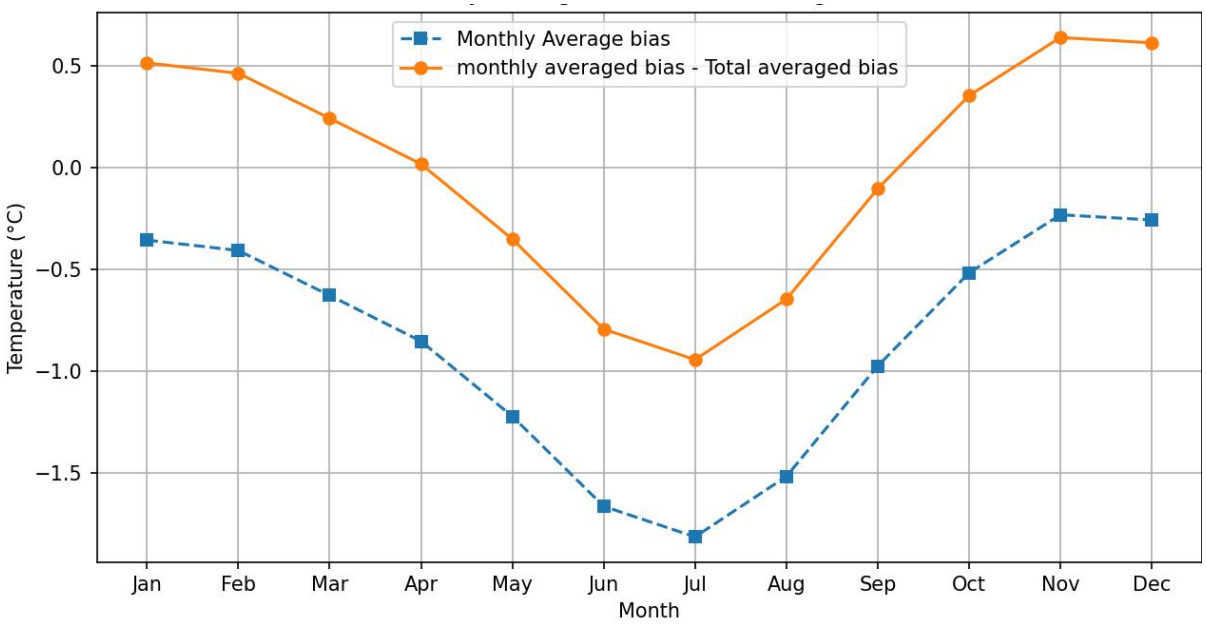


Figure 2-11 Monthly average bias and the difference between it and the total averaged bias

Bias Correction Map 3: As suggested by references [11, 21], considering that SST distributions are generally more similar the closer they are in time, the bias correction map for the day prior is calculated and then added to the current prediction map.

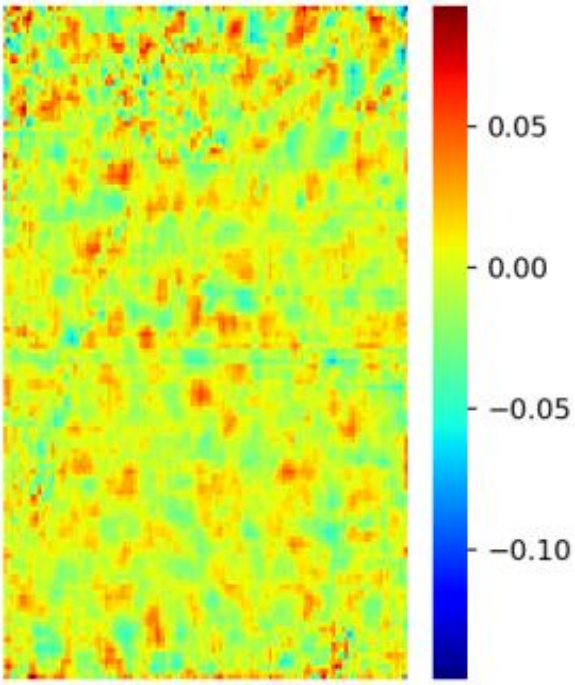


Figure 2-12 The Bias Correction Map 3 comes from the bias of the preframe

To validate the effectiveness of the three bias correction maps, they were applied during the post-processing stage of the model's predictions. The impact of these corrections was quantified by calculating the errors on the validation set, as detailed in Table 3-3.

Table 3-3 Effect comparison among different Bias correction map

	No Bias Correction Map	Bias Correction Map 1	Bias Correction Map 2	Bias Correction Map 3
MSE (°C²)	0.0566	0.0563	0.056	0.0531
MAE (℃)	0.182	0.181	0.181	0.1770
RMSE (℃)	0.2375	0.237	0.2365	0.2302
Bias (℃)	0.0016	-0.00149	-0.00145	-0.0006

3 Results and analysis

3.1 Error Measurement Metrics

For this sea surface temperature (SST) forecasting task, Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Bias were selected as the evaluation metrics. Each of these metrics reflects the nature of prediction errors and model biases from different perspectives. The combined use of these four metrics helps provide a comprehensive understanding of the model's accuracy, robustness, and systematic biases, thereby offering more insights for model optimization and improvement.

Suppose 'I' represents the predicted SST distribution map, and 'K' represents the ideal prediction output, i.e., the next frame image. Here, 'N' and 'M' denote the width and height of the temperature map matrix, respectively. A smaller MSE value indicates greater similarity between the images.

3.1.1 Mean Squared Error (MSE)

This is calculated by squaring the differences between predicted and actual values for each grid point, summing these squares, and then averaging the result. MSE is more sensitive to large errors because the errors are squared. This helps to identify outliers or shortcomings in model performance that lead to large errors.

$$MSE = \frac{1}{M \bullet N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i,j) - K(i,j)]^2$$

3.1.2 Mean Absolute Error (MAE)

MAE is calculated by taking the absolute value of the difference between predicted and actual values for each grid point, summing these absolute values, and then averaging. MAE provides a direct perception of the size of the errors and is more robust, as it is not as sensitive to large errors as MSE. This is useful for assessing the average performance of the model under typical conditions.

$$MAE = \frac{1}{M \bullet N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} |I(i,j) - K(i,j)|$$

3.1.3 Root Mean Squared Error (RMSE)

RMSE, Also known as the standard deviation of regression systems, RMSE is the square root of MSE. RMSE reduces the penalty for large errors relative to MSE and maintains the error metric on the same scale as the original data, making it easier to interpret.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{M \cdot N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i,j) - K(i,j)]^2}$$

3.1.4 Bias

Bias is calculated by averaging the differences between predicted and actual values across all grid points. Bias indicates the average deviation of model predictions, providing the most intuitive assessment of whether the model systematically overestimates or underestimates the actual values. This is particularly useful for diagnosing if there is a systematic error in the model, corresponding to the bias correction map added in the post-processing stage.

$$Bias = \frac{1}{M \cdot N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} |I(i,j) - K(i,j)|$$

3.2 Prediction comparison with other models across 3 regions

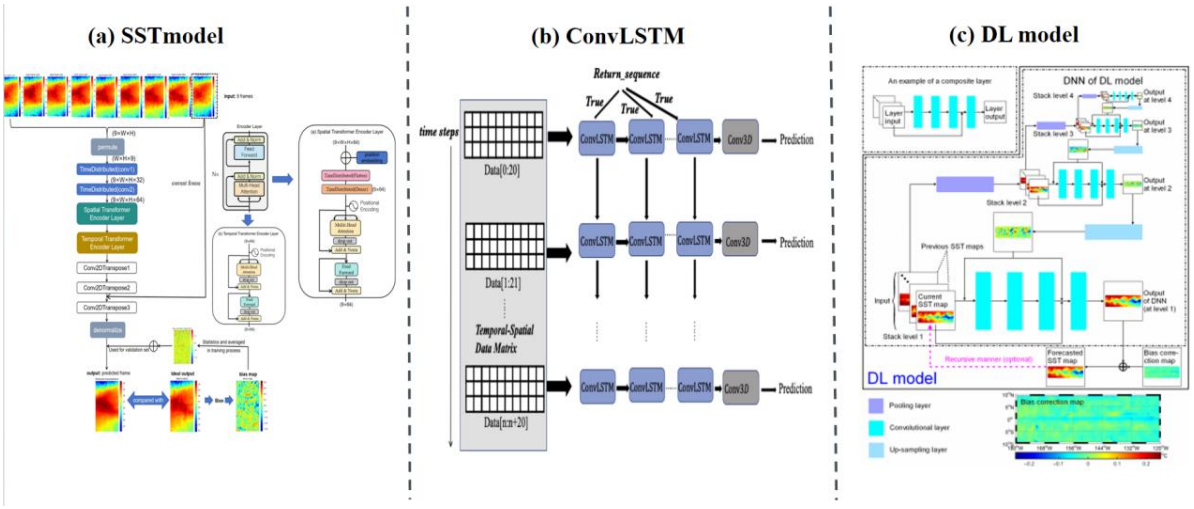


Figure 3-1 3 SST prediction models used for effect comparison

To validate the effectiveness of the proposed model (SSTmodel), the ConvLSTM structure and the 'DL' model from [11], both of which were proved good performance in

similar SST spatio-temporal sequence prediction tasks, were published in high quality publications in recent years.

these tables corresponds to comparison results among 3 models when applied to predict in 3 regions. Among them, the SSTmodel shows the best performance in all metrics when applied to different regions, with a bias of approximately ±0.001 ℃ and an RMSE not exceeding 0.24 ℃.

Table 3-1 The prediction error using 3 models in region 1

	SSTmodel	ConvLSTM	DL model
MSE (°C²)	0.0578	0.2668	0.144026
MAE (℃)	0.1787	0.389	0.2838
RMSE (℃)	0.240	0.5161	0.366
Bias (℃)	0.00101	-0.00426	-0.00134

Table 3-2 The prediction error using 3 models in region 2

	SSTmodel	ConvLSTM	DL model
MSE (°C²)	0.053	0.1145	0.083
MAE (℃)	0.177	0.2431	0.2074
RMSE (℃)	0.230	0.3231	0.286
Bias (℃)	0.0006	-0.00149	-0.02398

Table 3-3 The prediction error using 3 models in region 3

	SSTmodel	ConvLSTM	DL model
MSE (°C²)	0.0446	0.076	0.0836
MAE (℃)	0.161	0.204	0.2072
RMSE (℃)	0.211	0.275	0.287
Bias (℃)	0.0011	-0.00215	-0.01669

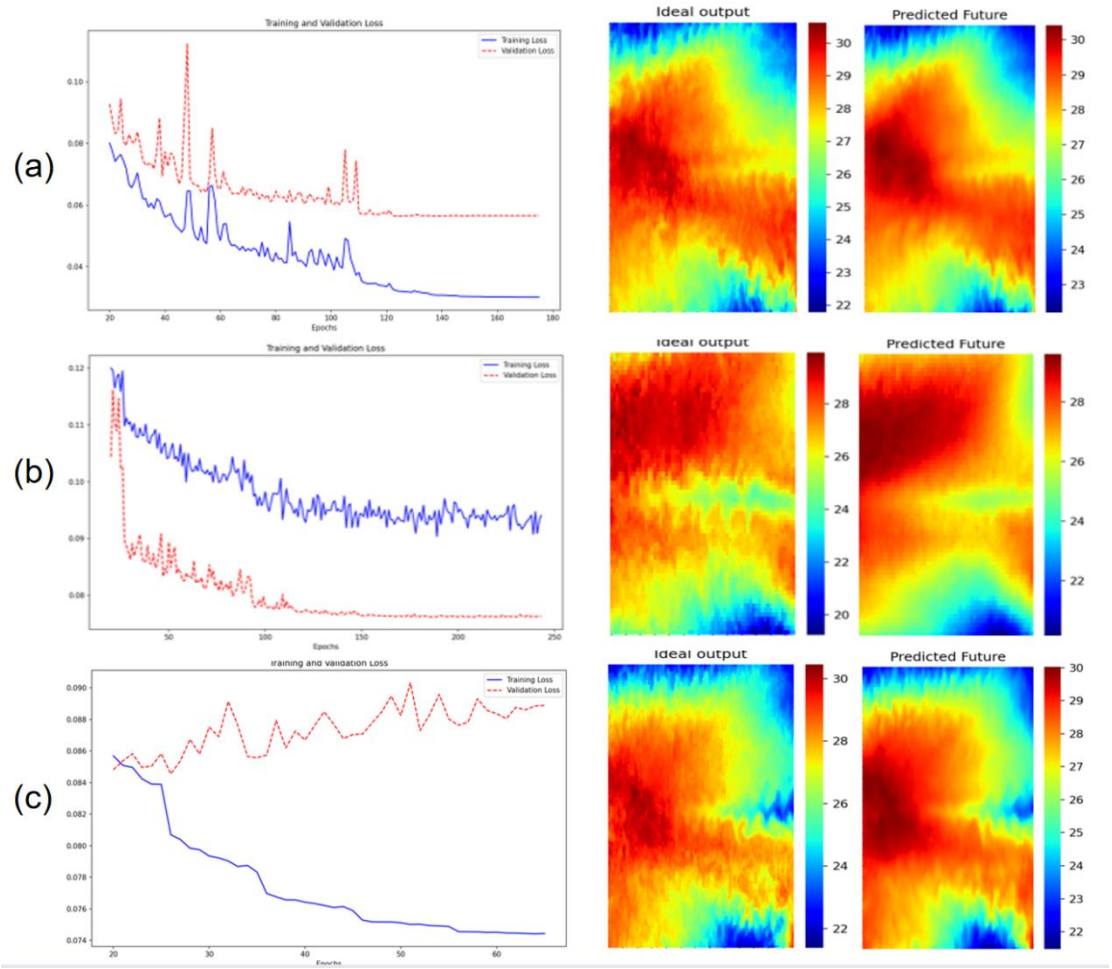


Figure 3-2 The Training results and outputs. (a) SSTmodel; (b) ConvLSTM and (c) DL model

3.3 Hardware resources used and analysis

Table 3-4 Effect of 3 models in region 3

	SSTmodel	ConvLSTM	DLmodel
running speed	121s/epoch	802s/epoch	21s/epoch
Loss (validation set, MSE)	0.0446°C²	0.076°C²	0.0836°C²
Loss (training set, MSE)	0.0298°C²	0.072°C²	0.0744°C²
convergence rate	130 epochs	150 epochs	60 epochs

In addressing the task of sea surface temperature (SST) prediction, three distinct deep learning models were analyzed and compared: the self-attention-based SSTmodel, the recurrent neural network-based ConvLSTM model, and the convolutional neural network-based 'DL model' [15]. These models exhibited significant differences in running speed, loss performance, and convergence speed when trained on CPUs, which primarily stem from their structural characteristics and data processing capabilities.

Firstly, from the perspective of running speed, the DL model significantly outperformed the other two models, requiring only 21 seconds per epoch. This advantage is mainly due to its simpler convolutional network structure [15], which is highly efficient in processing image and spatial data, allowing for rapid data handling. Consequently, it has the fastest single epoch running time among three models. The ConvLSTM model, due to its recursive nature, requires sequential processing of time steps in the LSTM part, which is particularly time-consuming on CPUs because it cannot be parallelized as easily as convolution operations. Hence, it has the slowest running speed, requiring 802 seconds per epoch. In contrast, the SSTmodel has a moderate speed, needing 121 seconds per epoch. Although the self-attention mechanism theoretically allows for parallel processing. However, self-attention calculations remain computationally intensive without GPU acceleration in practice. Calculating weights between all data points is particularly time-consuming on large datasets.

Secondly, in terms of loss performance, the SSTmodel shows superior learning and generalization capabilities on both the training and validation sets, with MSE losses of 0.0298 °C²and 0.0446℃, respectively. This performance is attributable to the self-attention mechanism that allows the model to capture complex relationships between any two points in the time series, regardless of their temporal distance. This capability enables the model to better understand and predict the dynamic changes in sea surface temperature, thereby achieving the lowest MSE losses on both the training and validation sets. In contrast, although the ConvLSTM can handle spatial and temporal dependencies in time series, its performance in capturing long-term dependencies may be limited by its recurrent structure, especially in long sequences and complex dynamics. And it has very low accuracy on high resolution, small range data of 64 * 64, and works better on the smoother datasets. The DL model, while being fast and exhibiting low losses in early training, suggests potential overfitting to the training data, leading to inferior generalization on the validation set compared to the other two models.

Finally, regarding convergence speed, the DL model converges rapidly after 60 epochs, possibly due to its simpler structure and rapid adaptation to specific training data. However, its MSE on the training and validation sets indicates that rapid convergence may compromise the model's generalization ability, resulting in overfitting. The SSTmodel requires more epochs to adjust its complex self-attention network parameters to achieve global optimality, with convergence occurring at 130 epochs. ConvLSTM converges the slowest, requiring 150 epochs. Although SSTmodel and ConvLSTM converge more slowly, they provide more robust predictive performance.

Besides, some methods were tried to improve the effect. For example, attempts were made to use date information as a feature for multi-feature input to further learn the seasonality and cyclicity of SST changes. However, the prediction effectiveness decreased. Therefore, more data collection on other parameters is needed to explore their usability and their correlation with SST changes.For the specific multivariate input model design, [7, 18] can be referred for modification.

In summary, the design philosophy and structural features of each model determine their performance in the task of predicting sea surface temperatures. The SSTmodel, with its self-attention mechanism, excels in generalization capability and predictive accuracy. Although its training and convergence speeds are slower than those of the DL model, considering the accuracy of predictions and the practical value of the model, SSTmodel offers a more superior solution in this scenario. Moreover, SSTmodel demonstrated the best prediction accuracy in all three regions, proving its excellent generalization ability, both globally and regional. Based on this, it can be further tried to apply to larger data sets.

3.4 Long-term prediction accuracy of SSTmodel

In long-term predictions, figure 3-3 is the global prediction error result of the SSTmodel across 250 days in 50 recursive time steps. Even in the prediction of the farthest 50th time step, the Bias is less than 0.0118°C.

The specific principle of using recursive time steps for long-term prediction is as follows: the model first uses the initial 9 frames (1st-9th frame) of input data to predict the first time step (10th frame). After completing the prediction for the first time step, the predicted result (10th frame) is used as final frame of the inputs for the next prediction. It is combined with the original input data from the previous 8 frames (2nd-9th frame) to form a new set of 9-frame inputs, which is then used to predict the next time step (11th frame). This process is repeated, with each new prediction result being recorded and combined with the previous 8 frames of input data, progressively moving forward until all 50 time steps have been predicted. The global loss of the SSTmodel at each recursive step were calculated from the forecasting errors of all samples at all grids, and then average result across all samples of the same time step.

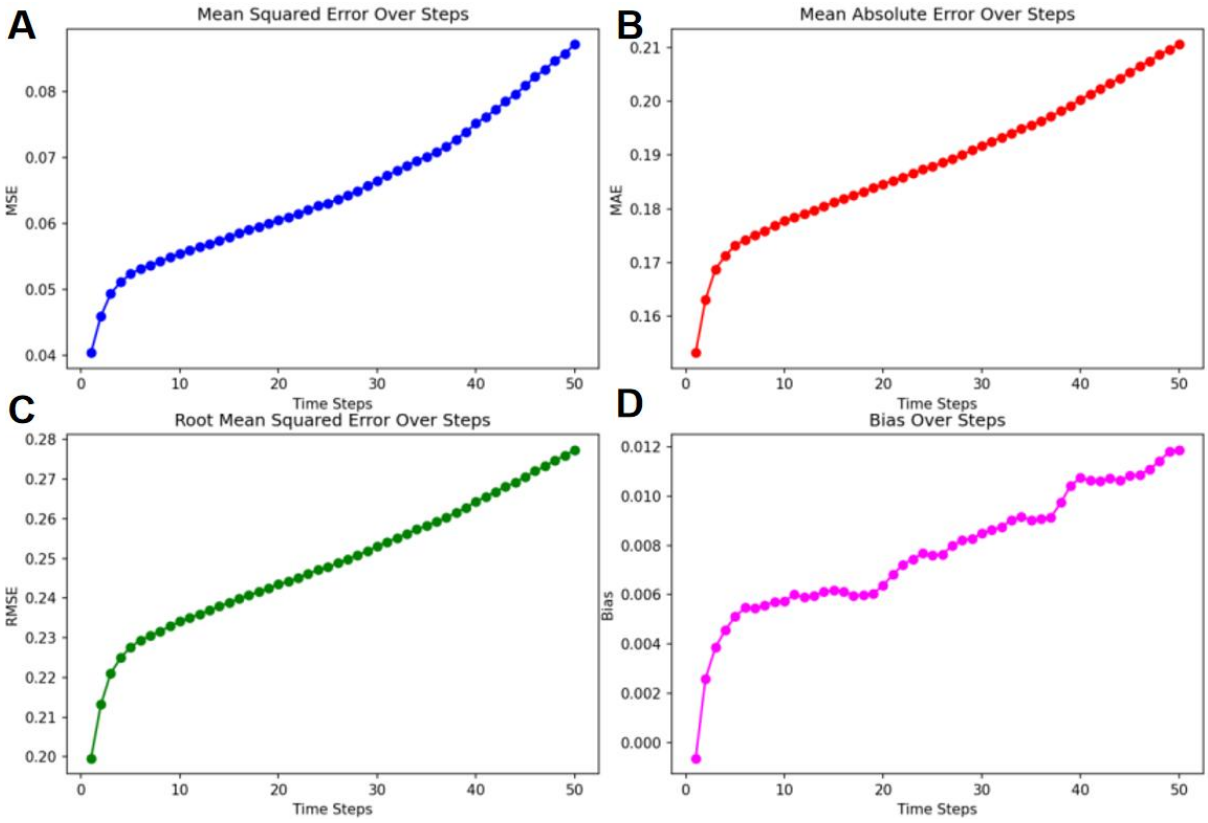


Figure 3-3 Trends of global loss of SSTmodel implemented recursively concerning the number of recursive steps. (A) MSE; (B) MAE; (C) RMSE and (D) Bias

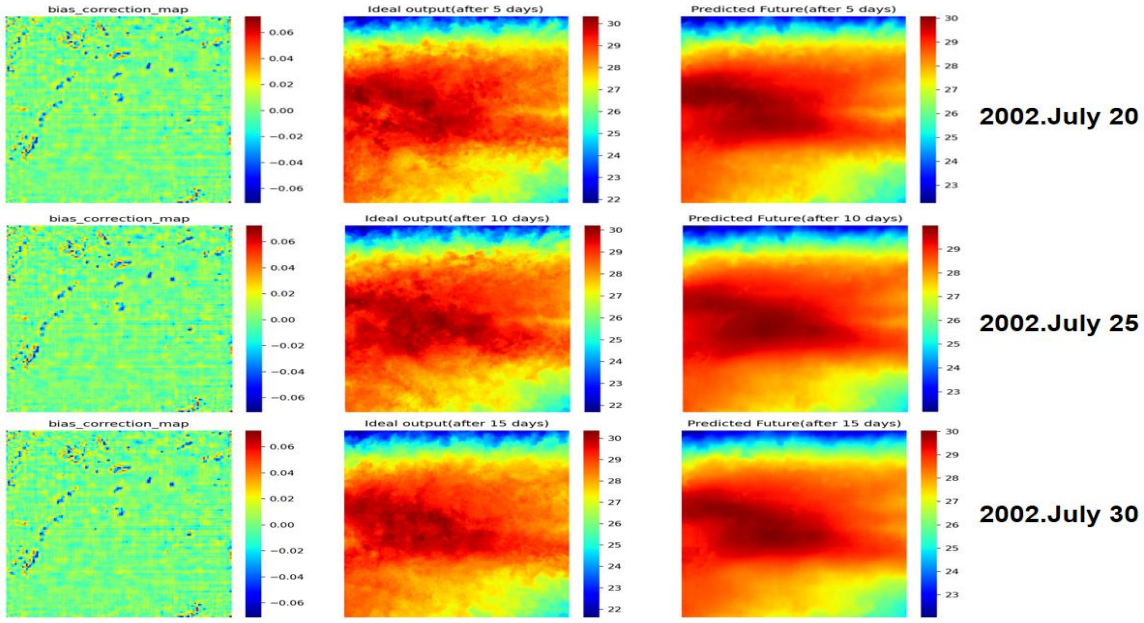


Figure 3-4 A specific example of predicted results of 3 recursive time steps

For example, using SST distribution maps from June 1, 2002, June 6, 2002, June 11, 2002, ..., up to June 11, 2002, a total of 9 frames, the model predicts the next 3 time steps of SST distribution as figure 3-4.

3.5 Niño 3.4 index prediction and identification

In figure 3-5, The blue line represents the actual value, and the red line represents the predicted value. It can be found that two lines fit well except when SST changed extremely, which usually correspond to the anomalously warm El Niño phenomenon and the anomalously cold La Niña phenomenon, which can be determined using the Niño 3.4 index. The Niño 3.4 index is defined as the sea surface temperature (SST) anomaly in the equatorial eastern Pacific region (specifically between 5°N-5°S, 170°W-120°W). The criteria are as follows: an El Niño event is declared when the SST anomaly in the Niño 3.4 region exceeds 0.5°C for at least five consecutive months, while a La Niña event is declared when the SST anomaly remains below -0.5°C for at least five consecutive months. These phenomena usually lead to the occurrence of extreme weather and have significant impacts on global climate [26].

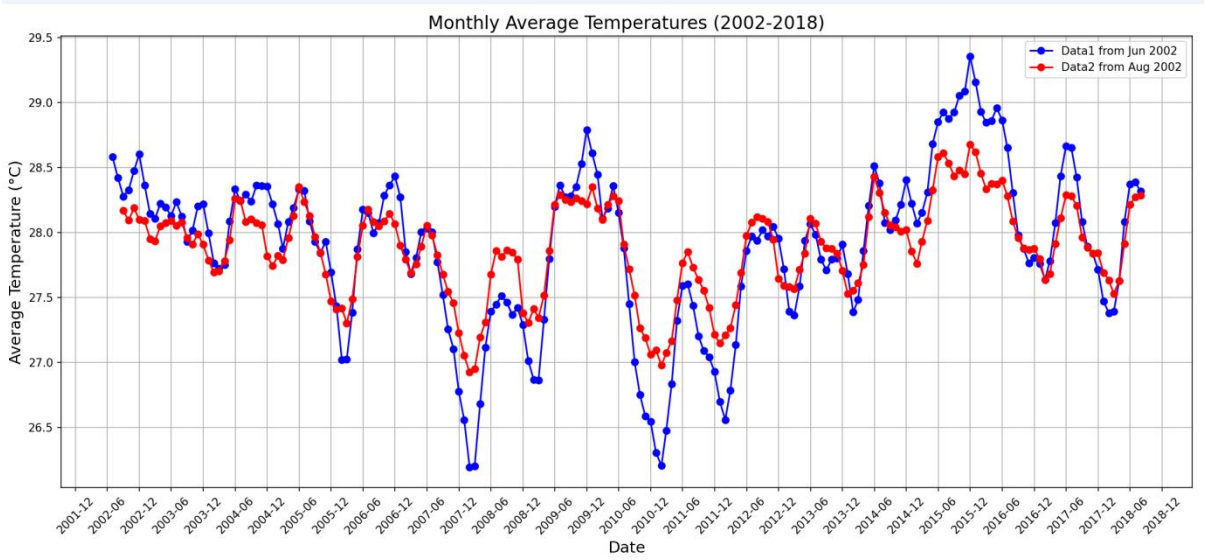


Figure 3-5Average bias between actual and prediction in each month

The left part in figure 3-5 is the Niño 3.4 index obtained by the predicted data; The right part in figure 3-6 is the actual Niño 3.4 index obtained from the NOAA website. They also fit well, from that, The El Niño and La Niña phenomena between 2007 and 2018 can be judged effectively.

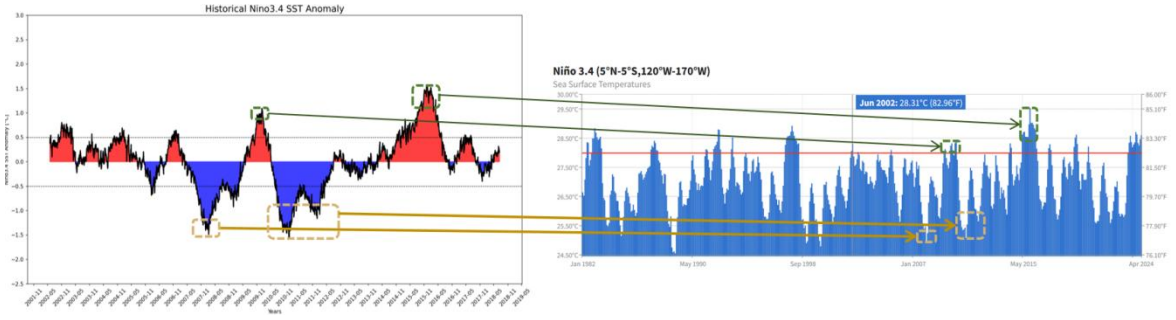


Figure 3-6 Niño 3.4 index among 2002 year to 2018 year meets the actual value of the period

3.6 Interactive visualization interface

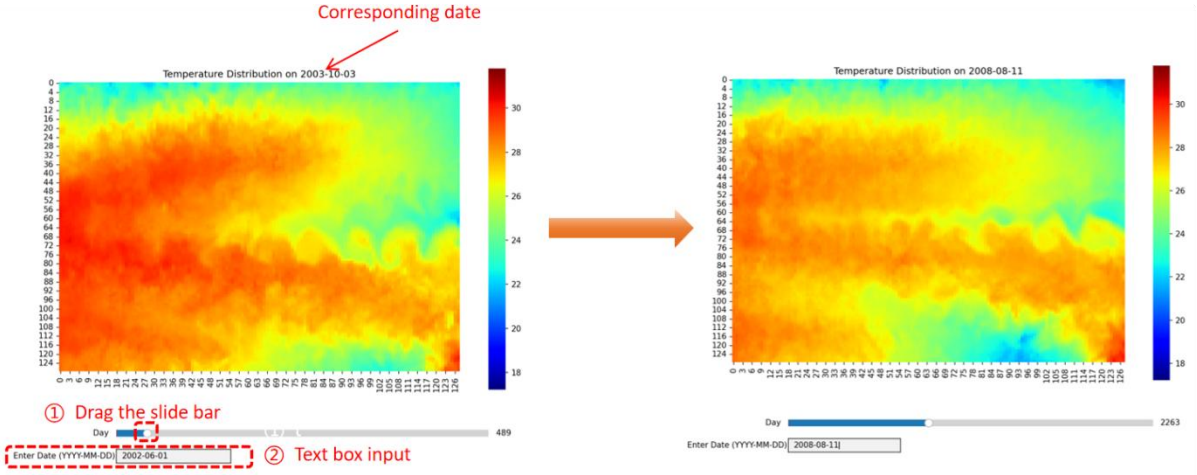


Figure 3-7 Interactive visualization interface

Finally, an interactive visualization interface was constructed to intuitively reflect the dynamic changes in SST distribution over time (the spatio-temporal evolution process of SST distribution). Users can jump to the corresponding date's SST distribution map by dragging the progress bar or typing the date in the text box. The whole actual samples and predicted images were collected for display, the implementation process includes the following steps: using ‘numpy.datetime64’ to determine the start date and create a date sequence corresponding to the length of the temperature data. The initial temperature distribution is displayed using ‘seaborn.heatmap’, with setting a specified color map and colors. Users can select the temperature matrix by day using a slider or quickly jump to a date using a text box. To achieve such dynamic updating of the heatmap, the ‘update’ and ‘submit’ functions are used to respond to slider and text box events. Finally, ‘plt.show()’ displays an interface containing the heatmap and interactive controls. This interface provides users with an intuitive way to view temperature distributions for different dates, helping to deepen their understanding of temperature changes over time.

4 Software and Hardware

In conducting research on Sea Surface Temperature (SST) prediction, this paper utilized a variety of computational tools and software libraries to support the entire process of data handling, model development, and result analysis. During the data processing phase, NumPy and Pandas were primarily used. NumPy provides extensive support for array and matrix operations, serving as a fundamental library for high-performance scientific computing. Pandas plays a role in data manipulation and preprocessing, particularly suitable for handling and analyzing time-series data. The model building and training phases relied mainly on TensorFlow and its high-level API, Keras. TensorFlow, a powerful open-source machine learning framework, supports a wide range of algorithm implementations and is especially suited for large-scale computations. Keras, as an interface for TensorFlow, offers a simpler method to construct and train deep learning models, making experimental procedures quicker and more flexible.

For visualization analysis and presentation of results, Matplotlib and Seaborn were employed in this study. Matplotlib, the most widely used plotting library in Python, is suitable for creating high-quality graphics. Seaborn, which builds on Matplotlib, provides a more advanced interface capable of producing attractive statistical charts. All these tools were used within the PyCharm integrated development environment, which offers excellent code editing, project management, and debugging functionalities, significantly enhancing development efficiency and code quality. The combined use of these tools successfully enabled effective SST prediction, demonstrating the powerful capabilities and flexibility of these software in handling complex scientific data.

Regarding hardware, the model's design optimization ensured efficient operation on Central Processing Units (CPUs). The total parameter size of the model is 65.68MB, far below 1GB, indicating low requirements for hardware memory. The model training took approximately 83 seconds per round, with total training duration kept under five hours, effectively demonstrating good efficiency and performance even on standard or low-end hardware configurations. This also showcases the feasibility of accomplishing complex tasks through model architecture optimization in resource-limited environments. Furthermore, the advantage of low memory requirements provides the potential for deploying and running the model on devices with limited resources, lowering technological barriers and enhancing the model’s practicality and universality, offering high feasibility and cost-effectiveness.

5 Conclusions and future work

5.1 Conclusions

This project successfully developed a spatiotemporal sequence prediction model for Sea Surface Temperature (SST), which efficiently combines Convolutional Neural Networks (CNN) and the Transformer architecture. Initially, the model leverages a multi-scale convolutional network to effectively capture the spatial features of SST data, providing high-quality features for subsequent self-attention layers. This enables the model to more accurately analyze and understand spatial dependencies within the data. The design of the model fully utilizes the characteristics of spatiotemporal data, with separate spatial and temporal encoder layers effectively capturing the dynamic changes in SST. Moreover, the model integrates a specific error correction mechanism (bias correction maps) that optimizes outputs by analyzing the deviations between predicted results and actual data, ensuring the accuracy and reliability of the predictions. Compared to traditional ConvLSTM and simple convolutional network-based DL models, the SSTmodel shows superior performance in all metrics when applied to different regions and demonstrates the high prediction accuracy and good generalization capability. Additionally, its characteristics of efficient and resource-saving (with the total parameter size of 65.68MB, which is very small for large image processing), High accuracy in long- term prediction and the accurate identification of the occurrence of El Niño and La Niña phenomena further prove its potential and applicability.

5.1.1 Main Achievements and Learning Outcomes

- **Model Performance:** Through a meticulously designed network structure, the model achieves high accuracy in SST prediction tasks, showcasing the potential of deep learning in the analysis of Earth science data.
- **Low Hardware Requirements:** Despite the complexity of the model, the total parameter size is controlled at 65.68MB, well below 1GB, demonstrating the model's good adaptability and low operational costs across different hardware setups.
- **Training Efficiency:** The model exhibits excellent training performance on standard CPUs, with each training round taking approximately 121 seconds, and the total training duration not exceeding five hours, validating the model's computational efficiency.
- **High prediction accuracy and good generalization capability:** high prediction accuracy when applied to SST prediction in 3 regions with different scales: Bias of approximately ±0.001℃ and RMSE not exceeding 0.24℃.

- **Long-term prediction ability:** SST images of the following 50 recursive time steps (across 250 days) has been predicted using recursive time-step prediction, the extremely low loss has been achieved: MSE < 0.0872°C²; MAE < 0.21℃; RMSE < 0.2773℃ and Bias < 0.0118℃ shows high precision forecasting of SSTmodel in both long and short term forecasting.
- **Niño 3.4 index calculation and abnormal phenomena identification:** The predicted Niño 3.4 index among 2007 year to 2018 year meets the actual trend of this period, which can be used to effectively judge and predict the occurrence of El Nino and La Nina phenomena.

5.1.2 Goal Assessment and difficulty

- **Goal Comparison**: The initial aim was to develop a model capable of accurately predicting sea surface temperature with real-time responsiveness. Although the model met the high accuracy in different regions and long-term prediction, even the effective judgement of El Niño and La Niña phenomena. There is room for improvement in real-time performance and multi-parameter integration, particularly in data processing and model response times.
- **Task Difficulty**: During model development, data preprocessing and feature extraction proved to be more complex than anticipated, especially when dealing with large volumes of historical data in different dimensions. Moreover, maintaining model generalizability while controlling computational resource usage was challenging. Various marine factors interact with each other [8], and many studies are now beginning to develop multi-parameter fusion models to process inputs from different data sources better and capture the interactions between different features to achieve more accurate predictions. This awaits further data collection and model optimization.

Through the execution and results of this project, a strategy to effectively combine CNNs and Transformers for processing spatiotemporal data was realized, highlighting the importance of model optimization in resource-limited situations. Future work could explore applying this model type to other similar environmental monitoring and prediction tasks, further optimizing the model structure to enhance processing speed and reduce computational resource demands.

5.2 Future work

This project has achieved preliminary results in the field of Sea Surface Temperature (SST) prediction and has also revealed some new research directions and possibilities for

improvement. Given that the SST prediction task is expected to further explore its potential for practical application, it is also necessary to explore the application effect in other more sea areas and recent real SST data. In addition, to further enhance the model's performance and practicality, the following are some feasible suggestions for future work:

5.2.1 Improving Model Generalization

The current model performs well on specific datasets, but its adaptability in new geographic regions or under different environmental conditions needs to be validated. Subsequent work could involve testing with more diverse datasets or developing new algorithms to enhance the model's generalization capabilities.

5.2.2 Deepening Model Interpretability

To better understand the basis of model predictions, future research could focus on enhancing model interpretability. This includes studying key factors in the model's decision-making process and exploring how to visualize these factors to help scientists and decision-makers better understand model outputs.

5.2.3 Exploring New Model Architectures

Considering the current architecture and functions of SSTmodel, the relevant literature was searched to further understand the research results of spatial and temporal sequence prediction. Based on it, the following plans for model improvement and research development are developed:

- The SSTmodel may benefit from integrating multistage attention mechanisms, which are described in detail in the article by Hu and Zheng on multivariate time series prediction [27]. This approach allows the model to better capture the influence of various non-predictive time series at different time stages, and thus to more accurately capture the key factors affecting SST.
- To enhance the performance of the SSTmodel in processing sea surface temperature data, Introducing a dynamic hypergraph structure and multi-view spatiotemporal convolution blocks [28] is planned to experiment with. The dynamic hypergraph structure, with its capability to represent multi-relational hyperedges, can effectively capture the complex dependencies within spatiotemporal data, thereby improving the model's adaptability to dynamic environmental changes and its accuracy in predictions. Concurrently, the multi-view spatiotemporal convolution blocks, by processing data across various temporal and spatial scales in parallel, can finely capture features from macro to micro levels, enhancing the model’s understanding of complex data patterns. The combination of these

two techniques will help the model more accurately predict and understand cross-regional influences, enhancing its expressive power and ability to handle complex systems.

- With the rapid development of machine learning and deep learning technologies, new model architectures continue to emerge. Future research might consider exploring innovative architectures such as Variational Autoencoders (VAEs) or Generative Adversarial Networks (GANs) to further enhance prediction accuracy and efficiency.

5.2.4 Developing Multi-Parameter Fusion Models

In SST prediction research, while SST itself is a key indicator, the variation in SST is influenced by various marine and atmospheric parameters [26], which are highly random and uncertain, such as Sea Surface Salinity (SSS), ocean current speed, sea surface wind speed and direction, and atmospheric pressure [8, 10, 29], and [8] shows that combining more surface parameters like wind speed, radiation and heat balance at the surface.But the data parameter used here is small, only a separate feature: SST, that is, using only past SST data for training.

Solution: To improve prediction accuracy, it is suggested that future research should implement more comprehensive multi-feature analysis by integrating multiple marine parameters. Future studies could integrate more environmental monitoring data, such as atmospheric temperature, salinity, and ocean currents, to build a comprehensive predictive model, which would help in a more holistic understanding and prediction of SST changes. If without changing the original dataset, the approach is to maximize the utilization of available information and explore additional dimensions. Firstly, considering the presence of islands can influence the surrounding oceanic climate, leading to uneven variations in SST across regions. For this dataset, apart from open sea areas, there are many islands in actually, which can be differentiated by using np.meshgrid( ) and globe.is_land( ) to create masks for land areas, effectively introducing an water or land variable. Secondly, taking into account the seasonal variations and their corresponding changes [8], the whole data can be cut into slices corresponding to different time periods for separate analysis. For a more comprehensive perspective, a dynamic approach can be integrated into modeling. Furthermore, combining a dynamic approach to construct ocean dynamic models along with relevant knowledge can enhance the analysis. Moreover, a trainable multimodal inversion scheme can be applied to reconstruct the sea surface dynamics effectively from multi-source [30] by satellite-derived observations.

5.2.5 Applied to the larger-sized image data

As the geographical area expands from Region 1 to Region 3, the error predicted by the SST model remains within a small range, demonstrating the SSTmodel's good adaptability to different resolutions and data volumes. Given that ocean systems flow interactively on a global scale, applying the model to larger SST distribution maps would be beneficial for learning more about temperature variation patterns, such as changes in ocean currents. Currently, to conserve hardware resources, a Dense structure is used in the model's prediction process to transform the feature map from (9, 64, 64, 64) to (9, 64), which may also result in the loss of information. From this perspective, it might be effective to progressively transform and reduce image sizes by stacking multiple layers of Dense structures, or by putting larger-sized inputs to multi-head attention layers directly to retain more information.

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